

Robust Nighttime Vehicle Localization Under Headlight Saturation via YOLO Detection and Actuation-Aware 3-D Tracking

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Abstract

Nighttime glare, scattering, and abrupt exposure changes can destabilize vehicle localization and produce incorrect adaptive driving beam (ADB) commands. This paper presents AuraBeam, an actuation-aware framework that couples a primary YOLO11n detector with a model-based temporal depth prior, adaptive six-state 3-D Kalman filtering, MBOX command generation, and an Arduino-compatible 8×8 LED Hardware-in-the-Loop layer. Legacy single detectors and ensembles are retained only as diagnostic comparisons; the main staged study uses one fixed detector. Downstream behavior is evaluated with GSSR and M-GSSR, which measure suppression fidelity after LED-grid projection, together with coverage and false-darkening metrics. In the historical five-case protocol, adaptive observation switching preserves the control cases and matches fixed observation in fog and lightning, but lowers occlusion GSSR from 77.78% to 55.56% in rain. A control-target filter subsequently raises D6 GSSR from 5.85% to 13.02%, reduces D6 FDR from 64.82% to 44.56%, and raises D7 8×8 M-GSSR from 26.10% to 44.49%. CPU timing further shows that ensembling adds at least 41.6% mean latency relative to its constituent single detectors, while every tested configuration misses the 33.3-ms deadline. The results show that robust ADB control depends on target selection and computational feasibility as well as detector and tracker quality.

Keywords: adaptive driving beam, nighttime vehicle localization, hardware-in-the-loop, Kalman filtering, glare suppression

1 Introduction

Nighttime crashes are disproportionately severe, and oncoming-headlight glare degrades visual performance [1, 2]. Adaptive Driving Beam (ADB) systems mitigate glare by suppressing only the beam region directed at another road user [3, 4]. Their camera-centric control loop is nevertheless vulnerable to blooming, flare, reflection, and exposure transients: unstable or missing detections can directly produce missing or misaligned LED commands [2, 5].

Nighttime-specific detectors and low-light trackers improve localization under sparse, high-intensity evidence [5–7], while sensor fusion improves robustness when vision is unreliable [8, 9]. However, these lines of work rarely address the ADB requirement to maintain a discrete suppression zone through short visual dropouts. Pixel-space metrics also do not reveal whether localization errors become residual glare or unnecessary darkening after LED-grid projection.

We therefore propose **AuraBeam**, a low-cost framework that combines a primary YOLO detector, a temporal Pseudo-Radar depth prior, adaptive six-state KF3D tracking, and mbox-based LED actuation. Historical detector ensembles are retained only as diagnostics. The system is evaluated through offline grid replay and an HIL-capable actuation layer using GSSR, an application-specific measure of suppression fidelity. The contributions are:

- an end-to-end primary-detector, depth-prior, KF3D, and LED-grid research stack;
- a fixed-versus-adaptive observation study covering controls, targeted degradation, and a rain limitation case;
- GSSR/M-GSSR evaluation that links localization to discrete actuation; and
- a control-target filter for reducing false darkening from oversized, stale, or excessive tracks.

The contribution is an actuation-aware engineering study rather than a general detector or MOT benchmark.

2 Background and Related Work

ADB has evolved from binary high-beam switching to spatial beam shaping driven by camera perception and adaptive modulation [3]. Production designs must also satisfy visibility and glare constraints such as ECE-R123 [4, 10]. Because the suppression mask is derived from image-plane detections, saturation and exposure changes can interrupt the command even when the optical actuator remains functional [2].

Nighttime detection relies on sparse light sources rather than stable texture, and blooming changes both target appearance and apparent extent [2, 5, 6]. Lighting-aware detection and low-light tracking address this regime [5, 6], but precision and

mAP do not measure whether a target remains correctly mapped to LED cells during dropout.

Kalman filtering provides a lightweight way to propagate state through missing observations [11]. Automotive fusion further shows that range or motion cues can complement camera semantics in darkness and adverse weather [8, 9]. AuraBeam adopts this principle in a constrained form: a temporal depth surrogate supports short ADB dropouts, without claiming equivalence to physical radar-camera fusion.

HIL, SIL, and VIL methods provide repeatable validation between simulation and on-road testing [12, 13]. Yet lighting studies usually emphasize functional or photometric verification, whereas vision studies stop at image-space metrics.

The remaining gap is a low-cost end-to-end protocol that sustains a camera-centric ADB command during glare-induced observation loss and evaluates the resulting discrete actuation. AuraBeam targets this intersection.

Camera-centric ADB therefore still lacks a tested response to glare-induced detection collapse. AuraBeam combines a temporal depth prior with adaptive Kalman filtering to maintain short-horizon actuation continuity and evaluates the result at the LED-grid level rather than only in pixel space.

Together, these strands motivate a low-cost, end-to-end study of glare-induced observation loss in camera-centric ADB, with localization continuity and discrete actuation evaluated in the same protocol.

3 System Methodology

3.1 System Overview

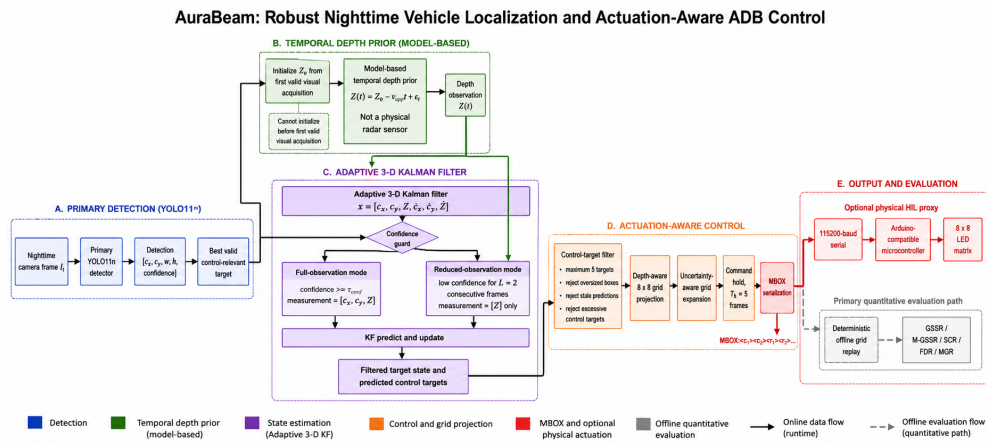


Figure 1 presents the complete AuraBeam pipeline rather than a comparison with a YOLO-only baseline. Module A applies the fixed primary YOLO11n detector

and selects valid control-relevant targets. Module B supplies a model-based temporal depth prior, and Module C combines the image-plane and depth observations through adaptive KF3D. Module D filters implausible control targets before depth-aware grid projection, uncertainty-aware expansion, five-frame command hold, and MBOX serialization. Module E separates the primary deterministic replay path from the optional physical HIL output. Table 1 summarizes the principal notation used throughout the paper.

Table 1: Main symbols and notations.

Symbol	Description
I_t	Input frame at time t
$\mathcal{M}$	Primary YOLO detector used by AuraBeam
$\mathcal{D}_t$	Detection set produced by $\mathcal{M}$ at time t
$b_j = [c_x, c_y, w, h]^\top$	Bounding box in center-width-height format
s_j	Confidence score of a detection
$\hat{b}, \hat{s}$	Selected control-relevant bounding box and confidence score
$\mathbf{d}_t$	Image-space observation vector at time t
$\mathbf{x}_k$	3-D Kalman filter state vector at step k
$\mathbf{F}$	State transition matrix
$\mathbf{Q}$	Process noise covariance matrix
$\mathbf{H}_{\text{full}}$	Observation matrix in full-observation mode
$\mathbf{H}_{\text{occ}}$	Observation matrix in occlusion-aware mode
$\mathbf{R}_{\text{full}}$	Measurement covariance in full-observation mode
$\mathbf{R}_{\text{occ}}$	Measurement covariance in occlusion-aware mode
$Z(t)$	Nominal Pseudo-Radar depth prior at time t
$\tilde{Z}(t)$	Noisy Pseudo-Radar depth observation used by the filter
$\mathcal{B}_t$	Suppression region on the LED grid at time t
mbox_t	Matrix-box command containing LED-grid column and row bounds
$\mathbf{G}_t$	Output LED suppression grid at time t
τ_{conf}	Confidence threshold for observation switching
T_h	Command hold-time window

3.2 Module A: Primary YOLO11n Detection

Module A applies the fixed primary YOLO11n detector to each nighttime camera frame I_t . After detector post-processing and class filtering, it produces the valid detection set

$$\mathcal{D}_t = \{(b_j, s_j)\}_{j=1}^{N_t}, \quad b_j = [c_x, c_y, w, h]^\top, \quad (1)$$

where b_j contains the image-space center, width, and height of detection j , and $s_j \in [0, 1]$ is its confidence score. Predictions that do not meet the detector confidence and class-validity requirements are discarded before tracking or control.

The best valid control-relevant target supplies the primary image observation and confidence value

$$\mathbf{d}_t = [c_x, c_y, w, h, \hat{s}]^\top \in \mathbb{R}^5. \quad (2)$$

The geometry in $\mathbf{d}_t$ initializes or updates the KF3D state, while $\hat{s}$ drives the confidence guard in Module C. The first valid target also initializes Z_0 for the temporal depth prior in Module B. Additional valid detections remain available as track candidates and are subsequently screened by the control-target filter before actuation.

3.3 Module B: Pseudo-Radar Depth Estimation

Pseudo-Radar is a lightweight temporal range prior, not a physical radar or LiDAR sensor. Its design is motivated by the value of night-time depth cues [7]; it supplies a longitudinal state component that can continue through brief image saturation.

For each initialized target, the range prior follows a short-horizon constant-velocity model:

$$Z(t) = Z_0 - v_{\text{app}}t, \quad (3)$$

where Z_0 is the initial range and v_{app} is the assumed approach speed. The implementation supports shared or per-track depth and adds a small zero-mean perturbation:

$$\tilde{Z}(t) = \max(0, Z(t) + \epsilon_t), \quad \epsilon_t \sim \mathcal{N}(0, \sigma_Z^2). \quad (4)$$

The low-noise setting favors continuity over precise dynamic ranging; the prior is not directly affected by frame-level detector confidence.

The approximation in Eq. (3) is accurate only when the relative motion is close to constant velocity. If the true longitudinal acceleration is $a_{\text{real}} \neq 0$, then the actual range evolves as

$$Z_{\text{true}}(t) = Z_0 - v_{\text{app}}t - \frac{1}{2}a_{\text{real}}t^2, \quad (5)$$

and the resulting depth estimation error is

$$\Delta Z(t) = Z(t) - Z_{\text{true}}(t) = -\frac{1}{2}a_{\text{real}}t^2. \quad (6)$$

The quadratic error growth limits Pseudo-Radar to short horizons; it is not a substitute for production radar, LiDAR, or calibrated stereo depth.

3.4 Module C: Adaptive 3-D Kalman Filter

3.4.1 State Representation and Motion Model

To integrate image-plane localization with the auxiliary depth signal, AuraBeam maintains a six-dimensional state vector

$$\mathbf{x}_k = [c_x, c_y, Z, \dot{c}_x, \dot{c}_y, \dot{Z}]^\top \in \mathbb{R}^6, \quad (7)$$

where (c_x, c_y) is the image centroid, Z is range, and the remaining terms are image-plane and longitudinal velocities. A constant-velocity model is used:

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad (8)$$

where Δt is the frame interval. The base process noise covariance is initialized as

$$\mathbf{Q} = \text{diag}(0.08, 0.08, 0.03, 0.03, 0.03, 0.02), \quad (9)$$

with the first three entries for position and the last three for velocity. The values favor temporal continuity over frame-level jitter, accepting modest smoothing lag to stabilize LED commands.

In the staged evaluation, confidence and prediction age can adapt the measurement and process variances within bounded limits; the state and Kalman recursion remain unchanged.

3.4.2 Observation Models

The filter has two observation modes selected by the confidence guard.

Mode 1: Full observation.

With sufficient confidence, the state uses image-plane coordinates and depth:

$$\mathbf{H}_{\text{full}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \quad (10)$$

with measurement vector

$$\mathbf{y}_k^{\text{full}} = [c_x, c_y, \tilde{Z}(t)]^\top. \quad (11)$$

The corresponding measurement noise covariance matrix is

$$\mathbf{R}_{\text{full}} = \text{diag}(30.0, 30.0, 0.5), \quad (12)$$

where the first two terms are in px^2 and the third in m^2 .

Mode 2: Occlusion-aware reduced observation.

When the best confidence is below τ_{conf} , or no valid target is available, for L consecutive frames, the filter updates only depth:

$$\mathbf{H}_{\text{occ}} = [0 \ 0 \ 1 \ 0 \ 0 \ 0], \quad (13)$$

with measurement vector

$$\mathbf{y}_k^{\text{occ}} = [\tilde{Z}(t)]. \quad (14)$$

The motion model propagates (c_x, c_y) while the depth prior anchors the state. The reduced covariance is one-dimensional:

$$\mathbf{R}_{\text{occ}} = [0.5]. \quad (15)$$

Thus the guard separates temporal smoothing for noisy detections from depth-anchored propagation during short image-plane failures.

3.4.3 Recursive Estimation

The prediction step is given by

$$\hat{\mathbf{x}}_k^- = \mathbf{F}\hat{\mathbf{x}}_{k-1}, \quad (16)$$

$$\mathbf{P}_k^- = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^\top + \mathbf{Q}, \quad (17)$$

where $\hat{\mathbf{x}}_k^-$ and $\mathbf{P}_k^-$ denote the prior state estimate and covariance. The Kalman gain is then computed as

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}^\top (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^\top + \mathbf{R})^{-1}, \quad (18)$$

followed by the update equations

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k (\mathbf{y}_k - \mathbf{H}\hat{\mathbf{x}}_k^-), \quad (19)$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_k^-, \quad (20)$$

where $(\mathbf{H}, \mathbf{R}, \mathbf{y}_k)$ are selected according to the current observation mode.

3.4.4 Adaptive Detect-then-Track Procedure

Algorithm 1 summarizes the per-frame detect–predict–update and grid-mapping loop.

3.5 Module D: Zone Mapping and Command Scheduling

Pixel-to-Grid Projection

The filtered centroid $(\hat{c}_x, \hat{c}_y)$ is projected onto the 8×8 LED grid as

$$g_{\text{col}} = \left\lfloor \frac{\hat{c}_x}{W_{\text{frame}}} \cdot 8 \right\rfloor, \quad g_{\text{row}} = \left\lfloor \frac{\hat{c}_y}{H_{\text{frame}}} \cdot 8 \right\rfloor, \quad (21)$$

where W_{frame} and H_{frame} are image dimensions. The controller then forms a depth-dependent region and serializes it as an mbox command; uncertainty-aware projection adds a bounded covariance margin [14, 15].

Depth-Dependent Suppression Policy

The prototype uses binary rather than calibrated dimming. Range $\hat{Z}$ determines whether a target is projected and the integer expansion shown in Table 2; targets beyond 100 m are omitted.

Algorithm 1 AuraBeam detect-then-track loop with adaptive observation switching

Require: Video stream $\{I_t\}$, primary detector $\mathcal{M}$, confidence threshold τ_{conf} , approach velocity v_{app}

Ensure: LED suppression grid $\mathbf{G}_t$ for each frame

- 1: Initialize $\hat{\mathbf{x}}_0$, $\mathbf{P}_0$, $\mathbf{Q}$, $\mathbf{R}_{\text{full}}$, and $\mathbf{R}_{\text{occ}}$
 - 2: Estimate initial range Z_0 from the first valid target acquisition
 - 3: **for** each frame $t = 1, 2, \dots$ **do**
 - 4: Run $\mathcal{M}$ on I_t to obtain $\mathcal{D}_t$
 - 5: Select the highest-confidence valid target to obtain $\mathbf{d}_t$; set $\hat{s}_t = 0$ if none exists
 - 6: Compute depth observation $\hat{Z}(t)$ from the Pseudo-Radar prior
 - 7: Predict $\hat{\mathbf{x}}_t^-$ and $\mathbf{P}_t^-$ using Eqs. (16) and (17)
 - 8: **if** $\hat{s}_t \geq \tau_{\text{conf}}$ or fewer than L consecutive low-confidence frames have occurred **then**
 - 9: Set $\mathbf{H} \leftarrow \mathbf{H}_{\text{full}}$, $\mathbf{R} \leftarrow \mathbf{R}_{\text{full}}$, and $\mathbf{y}_t \leftarrow \mathbf{y}_t^{\text{full}}$
 - 10: **else**
 - 11: Set $\mathbf{H} \leftarrow \mathbf{H}_{\text{occ}}$, $\mathbf{R} \leftarrow \mathbf{R}_{\text{occ}}$, and $\mathbf{y}_t \leftarrow \mathbf{y}_t^{\text{occ}}$
 - 12: **end if**
 - 13: Update $\hat{\mathbf{x}}_t$ and $\mathbf{P}_t$ using Eqs. (18)–(20)
 - 14: Map $(\hat{c}_x, \hat{c}_y, \hat{Z})$ to an mbox command and rasterize it as $\mathbf{G}_t$
 - 15: Transmit the mbox command to the actuation layer
 - 16: **end for**
-

Table 2: Depth-dependent glare suppression policy.

Estimated range	Control action	Expansion
$\hat{Z} > 100$ m	Target omitted	–
$60 \leq \hat{Z} \leq 100$ m	Binary suppression around centroid	0 cells
$30 \leq \hat{Z} < 60$ m	Binary suppression around centroid	1 cell
$\hat{Z} < 30$ m	Binary suppression around centroid	2 cells

Command Hold-Time Scheduling

The implemented scheduler is a gap-hold mechanism rather than a union over all recently suppressed cells. Let C_t^{raw} be the command produced by the current frame and let C_t be the transmitted command. A non-empty current command replaces the previous command immediately; if the current frame produces no command, the last non-empty command is held for at most T_h frames:

$$C_t = \begin{cases} C_t^{\text{raw}}, & C_t^{\text{raw}} \neq \emptyset, \\ C_{t_{\text{last}}}^{\text{raw}}, & C_t^{\text{raw}} = \emptyset \wedge t - t_{\text{last}} \leq T_h, \\ \emptyset, & \text{otherwise.} \end{cases} \quad (22)$$

where t_{last} is the latest non-empty command frame. The historical setting is $T_h = 5$ frames (about 167 ms at 30 Hz). The scheduler bridges empty gaps but does not union old cells with a new command.

Control-Target Filtering

Before projection, the control interface removes oversized, stale, or excessive targets that can cause broad false-darkening commands. The fixed thresholds are listed in Table 3; detector outputs are otherwise unchanged.

Table 3: Control-target filter used in the filtered staged evaluation.

Parameter	Value	Purpose
<code>max_control_targets</code>	5	Limits the number of targets passed to LED-grid control.
<code>max_control_box_area_ratio</code>	0.12	Rejects boxes whose area is implausibly large relative to the frame.
<code>max_control_box_width_ratio</code>	0.35	Rejects boxes that are too wide to be reliable control targets.
<code>max_control_box_height_ratio</code>	0.45	Rejects boxes that are too tall to be reliable control targets.
<code>max_control_prediction_age</code>	10	Rejects prediction-only targets that have become stale.

3.6 Module E: Hardware Actuation Layer

The hardware layer converts each mbox command into an LED-grid state. The host sends the packet over serial and the microcontroller rasterizes it independently of perception.

Communication from the host computer uses a 115200-baud serial link with the packet format

`MBOX:<c1><c2><r1><r2>[<c1><c2><r1><r2>...]`

Each four-character group stores column and row bounds; the clear command is `MBOX:-1-1-1-1`. The direct row-column firmware loop provides a physical actuation proxy, not calibrated optical power, beam shape, or regulatory photometry. At 115200 baud with 8-N-1 framing, a packet containing n boxes has $N_b = 6 + 4n$ transmitted bytes, including the prefix and newline. Its nominal on-wire time is therefore

$$T_{\text{serial}}(n) = \frac{10(6 + 4n)}{115200}, \quad (23)$$

which ranges from 0.87 ms for one box to 2.26 ms for the configured maximum of five boxes; the 14-byte clear packet requires 1.22 ms. The firmware scans eight rows with 150 μs illumination and 40 μs blanking per row, giving a nominal full-matrix scan period of

$$T_{\text{scan}} = 8(150 + 40) \mu\text{s} = 1.52 \text{ ms}, \quad (24)$$

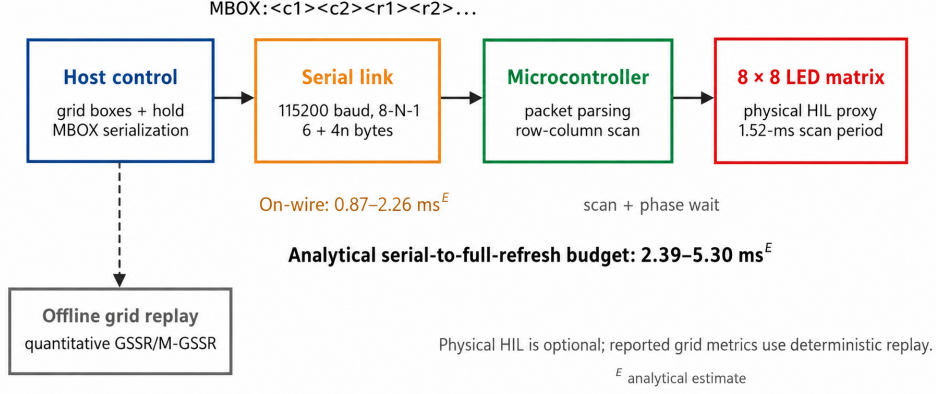


Fig. 2: MBOX actuation path and latency accounting. The host serializes grid boxes for either deterministic offline replay or optional transmission to the microcontroller and 8×8 LED matrix. Superscript E denotes an analytical timing estimate rather than a direct hardware measurement.

before digital-I/O and parsing overhead. Because serial input is polled once per scan loop, command arrival can add up to one scan-period phase wait. The resulting analytical serial-to-full-refresh budget is approximately 2.39–5.30 ms. This is a protocol/firmware bound rather than an ACK-based hardware latency measurement.

Figure 2 separates the deterministic replay used for quantitative grid metrics from the optional physical HIL path. This distinction prevents the analytical serial/LED budget from being interpreted as a measured photometric response.

AuraBeam therefore combines primary detection, a temporal depth surrogate, adaptive KF3D, and discrete actuation in an HIL-capable prototype with deterministic replay.

4 Evaluation Protocol

The experiments comprise three non-interchangeable tracks: (i) the main D0–D7 single-detector study, (ii) historical B/C ablations used to explain design choices, and (iii) external tracker comparisons. Results are compared only within a track because the detector checkpoint, frame subset, and metric aggregation are not identical across tracks.

4.1 Protocol Design and Implementation

The main study fixes the detector to the primary **best11n** checkpoint and uses adaptive KF3D tracking, the temporal depth prior, five-frame command hold, and mbox actuation. D0–D6 use the mountain, urban, rain, and fog scenarios; D7 freezes D6 predictions and replays the designated lightning validation scenario within the same

Golden Test Set (not an independent held-out test set) at 8×8 , 16×16 , and 32×32 resolutions. Table 4 defines the incremental stages.

Table 4: Main D0–D7 staged protocol. Each row adds one change to the preceding stage.

Stage	Configuration change
D0	KF3D baseline with default fixed noise, distance association, shared depth, and basic projection.
D1	D0 plus diagnostic logging; control behavior is unchanged.
D2	D1 with configured fixed process and measurement noise.
D3	D2 with confidence- and innovation-adaptive noise.
D4	D3 with multi-cue association and control-target filtering.
D5	D4 with per-track rather than shared depth.
D6	D5 with uncertainty-aware depth-centered projection.
D7	Frozen D6 prediction replay on the validation scenario at three grid resolutions.

The historical B/C track uses the archived weighted-NMS detector and the same five-scenario set. B1–B4 compare raw detection, KF2D, fixed-observation KF3D, and adaptive KF3D; C1–C4 isolate direct versus held commands and fixed versus adaptive observation. The archived B4/C4 runs use the recorded **best_conf** policy: two consecutive frames below $\tau_{\text{conf}} = 0.35$ activate depth-only updating. The interactive primary-detector runtime additionally treats a frame with no valid detection as low confidence. These rows explain the evolution of AuraBeam but are not results for the final primary detector. ByteTrack and BoT-SORT are evaluated separately with that detector.

The interactive runtime matches the main operating policy: primary detection, the same confidence guard, depth-dependent binary projection, and five-frame command hold. The staged evaluator additionally enables the D4–D6 modules in Table 4. All code runs in Python; the optional Arduino accepts the same mbox packets, whereas reported actuation metrics use offline grid replay. Experiments ran on an HP Victus with an AMD RyzenTM 5 7535HS without scenario-specific retuning. The dedicated latency benchmark uses CPU inference on this machine; accelerator results are not mixed into its runtime tables. Historical B/C configurations use five identical replays; D-stage results average five fixed seeds. These repeats/seeds assess reproducibility and configuration sensitivity, not independent stochastic subjects.

4.2 Evaluation Metrics

Actuation metrics are primary because AuraBeam outputs a discrete suppression command. Detector, trajectory, and runtime metrics are supporting diagnostics.

4.2.1 Actuation-Level Metrics

Let $\hat{\mathcal{B}}_t$ and $\mathcal{B}_t^*$ be the predicted and ground-truth suppressed-cell sets at frame t . Their grid IoU is

$$\text{IoU}_{\text{grid}}(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (25)$$

GSSR is the fraction of evaluated frames whose grid IoU reaches τ_g :

$$\text{GSSR} = \frac{1}{|\Omega|} \sum_{t \in \Omega} \mathbf{1}[\text{IoU}_{\text{grid}}(\hat{\mathcal{B}}_t, \mathcal{B}_t^*) \geq \tau_g], \quad (26)$$

where $\tau_g = 0.5$. In the historical B3/B4 track, Ω is the full sequence for controls and the annotated degradation interval otherwise.

M-GSSR applies the same condition after frozen prediction logs are reprojected at grid resolution r :

$$\text{M-GSSR}_r = \frac{1}{|\Omega_r|} \sum_{t \in \Omega_r} \mathbf{1}[\text{IoU}_{\text{grid}}(\hat{\mathcal{B}}_t^{(r)}, \mathcal{B}_t^{*(r)}) \geq \tau_g], \quad (27)$$

D7 reports 8×8 , 16×16 , and 32×32 results. Because D7 changes both grid resolution and frame set, its M-GSSR values are not compared numerically with historical GSSR.

SCR and grid-level FDR distinguish missed coverage from unnecessary darkening:

$$\text{SCR} = \frac{1}{|\Omega_+|} \sum_{t \in \Omega_+} \frac{|\hat{\mathcal{B}}_t \cap \mathcal{B}_t^*|}{|\mathcal{B}_t^*|}, \quad (28)$$

$$\text{FDR}_{\text{grid}} = \frac{1}{|\Omega_p|} \sum_{t \in \Omega_p} \frac{|\hat{\mathcal{B}}_t \setminus \mathcal{B}_t^*|}{|\hat{\mathcal{B}}_t|}. \quad (29)$$

Here Ω_+ and Ω_p contain frames with non-empty ground truth and prediction, respectively. Historical B/C tables use a frame-level false-darkening indicator, whereas D-stage tables use Eq. (29); the two forms are therefore reported within, not across, tracks. For empty unions, empty ground-truth masks, or empty prediction masks, the implementation assigns IoU, SCR, or FDR equal to zero, respectively.

Missed Glare Rate (MGR) is used as a simple failure-count metric:

$$\text{MGR} = \frac{1}{|\Omega_+|} \sum_{t: \mathcal{B}_t^* \neq \emptyset} \mathbf{1}[\hat{\mathcal{B}}_t = \emptyset]. \quad (30)$$

MGR captures complete command loss but not spatial misalignment; GSSR/M-GSSR therefore remain primary.

4.2.2 Supporting Metrics

Precision, recall, mAP@0.5, and mAP50-95 characterize detection. Centroid RMSE and jitter reduction (JR%) characterize trajectory accuracy and smoothness; latency or FPS characterizes runtime. IDF1 is omitted because the COCO annotations lack persistent identities, so no MOT identity-performance claim is made.

4.2.3 Latency and Deadline Metrics

Runtime is measured per frame with a monotonic high-resolution clock at four boundaries: input capture/decode, YOLO inference including preprocessing and post-processing, tracking plus mbox control generation, and the complete software control path. The latter begins immediately before frame acquisition and ends when the held mbox command is available; offline metric computation is excluded. For latency samples $\{T_t\}_{t=1}^N$, we report the population mean and standard deviation, the linearly interpolated 95th percentile, the maximum, and the deadline-miss rate

$$\text{DMR}_{33.3} = \frac{1}{N} \sum_{t=1}^N \mathbf{1}[T_t > 33.3 \text{ ms}], \quad (31)$$

where 33.3 ms is the frame budget for 30 FPS. The dedicated benchmark runs the five detector settings on the same 953 annotated frames. Five warm-up frames per scenario are excluded from latency statistics, leaving $N = 928$ steady-state samples per setting; GSSR still uses all 953 frames. Since the source is an offline video, the capture term measures file decode and frame transfer, not physical camera exposure or USB transport. Serial and LED timing are not included in the measured software path and are added only as the analytical budget from Eqs. (23)–(24).

4.3 Dataset, Scenarios, and Baseline Configurations

The Golden Test Set contains 953 manually annotated frames not used for detector training. Roboflow bounding boxes are projected to LED-cell ground truth using Eq. (21). The two control sequences test whether switching harms nominal behavior; fog and lightning provide sustained and brief targeted degradation; rain is a pre-declared limitation case. Table 5 contains the complete scenario definition, avoiding scenario selection after observing the results. The annotated degradation intervals are rain 117–125, lightning 25–33, and fog 33–56, 137–157, and 188–199 (zero-based video-frame indices).

Table 5: Golden Test Set scenarios and evaluation roles.

Scenario	Role	Frames	Primary difficulty
Mountain curve at night (night_1)	Negative control	76	Curved road and low visibility
Urban night with street lighting (norm_1)	Negative control	204	Street-light distractors
Rainy night (rain_1)	Limitation case	219	Reflections and lens droplets
Foggy night (fogging_1)	Targeted degradation	215	Bloom and diffuse glare
Lightning disturbance at night (thunder_1)	Targeted degradation / D7 validation	239	Abrupt exposure transient
Total		953	



Fig. 3: Representative Golden Test Set frames. (a) Mountain curve at night; (b) Urban night with street lighting; (c–d) Rainy night; (e) Foggy night; (f–g) Lightning disturbance at night.

Within each track, post-processing and actuation settings are fixed across scenarios. Scenario-wise results are retained because rain, scattering, and exposure transients produce different failure mechanisms; no test-time scenario-specific threshold tuning is used.

5 Experimental Results

Results are organized by evaluation question. We first compare detector choice and computational cost, then report the main single-detector D0–D7 control study, external online trackers, and the five-case boundary of adaptive observation switching. Archived B/C ablations and paired direction checks are reported separately in Appendix A so their legacy detector and aggregation protocol are not mixed with the main results.

5.1 Main Detector Pipeline and Historical Detector Comparison

Table 6 compares detector settings under the same five-scenario grid protocol. Only the primary checkpoint feeds the staged M-GSSR pipeline; the other rows are historical diagnostics.

Table 6: Main primary detector compared with historical detector baselines under the grid-level protocol. Actuation values are five-scenario means; runtime is reported separately in Table 7.

Detector setting	GSSR	SCR	FDR	Grid IoU	Role in this paper
Main primary YOLO11n checkpoint	7.31	53.71	60.84	0.138	Used in the staged M-GSSR pipeline
Historical YOLOv5n single detector	12.97	56.34	50.11	0.178	Detector diagnostic baseline
Historical YOLOv8 single detector	7.58	72.52	67.96	0.154	Detector diagnostic baseline
Historical NMS ensemble	6.78	69.53	71.99	0.134	Historical ensemble baseline
Historical weighted-NMS ensemble	7.18	69.11	72.44	0.129	Historical ensemble baseline

The primary checkpoint slightly exceeds the historical weighted ensemble in GSSR (7.31% vs. 7.18%) and has lower FDR (60.84% vs. 72.44%). The YOLOv5 diagnostic row has higher detector-only GSSR, but detector ranking is not the main claim: downstream control remains the bottleneck.

5.2 Latency, Deadline Compliance, and Computational Trade-off

Table 7 reports the dedicated CPU-only latency benchmark. Every configuration misses the 33.3-ms deadline on every measured frame, so the prototype is not a 30-FPS real-time implementation on the tested CPU. The percentile and maximum columns also show why mean latency alone is insufficient: the main YOLO11n checkpoint has a 248.38-ms mean but a 436.11-ms p95 and a 3503.88-ms maximum outlier.

Figure 4 visualizes the same computational trade-off. The horizontal segments extend from the mean to p95 latency and the hollow diamonds mark maxima. Both ensembles are slower than their constituent single detectors without a corresponding GSSR gain.

Table 8 decomposes the same path. Detector inference dominates all configurations; tracking and mbox generation remain below 2.4 ms on average. The final column adds the 2.39–5.30-ms analytical serial/LED budget to the measured software mean. It is therefore an estimated camera-input-to-full-grid-update range, not a direct photodiode or ACK measurement.

Table 7: Steady-state software-control latency and actuation trade-off on the local CPU benchmark. Latencies are in ms; GSSR uses all 953 frames, while latency statistics use 928 frames after warm-up removal.

Detector setting	Mean	Std.	P95	Max.	DMR	FPS	GSSR
Main YOLO11n	248.38	173.38	436.11	3503.88	100.00%	4.03	5.88%
Legacy YOLOv5n	156.03	45.91	213.90	770.65	100.00%	6.41	14.38%
Legacy YOLOv8	322.28	17.94	351.86	522.07	100.00%	3.10	9.13%
Legacy NMS ensemble	456.50	44.85	529.42	887.64	100.00%	2.19	8.29%
Legacy weighted ensemble	465.30	62.04	557.67	1269.59	100.00%	2.15	8.81%

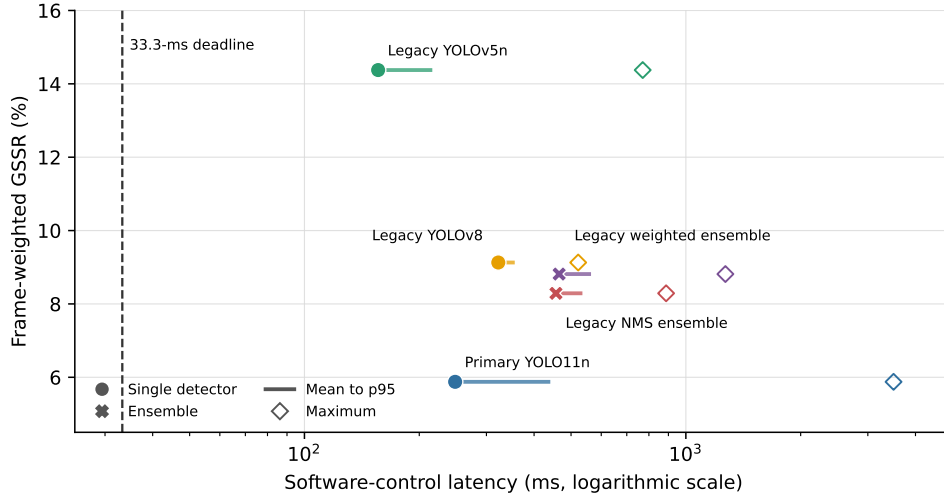


Fig. 4: Observed GSSR–latency trade-off on the CPU benchmark. Filled markers show mean software-control latency, horizontal segments end at p95, and hollow diamonds show maximum latency. The vertical line is the 33.3-ms frame budget.

Projected Deployment Latency

To translate the offline benchmark into a deployment-oriented engineering budget, we use the sequential, non-overlapped approximation

$$T_{\text{deploy}} = T_{\text{camera}} + T_{\text{detector}} + T_{\text{track/control}} + T_{\text{serial/LED}}. \quad (32)$$

Table 9 evaluates Eq. (32) under explicit assumptions. The 10–33.3-ms camera interval is an engineering envelope for exposure, sensor readout, transfer, and up to one frame of buffering at 30 Hz; it is not measured from a selected automotive camera. The 15–25-ms accelerated-detector interval is likewise a target envelope for a future GPU or edge accelerator, not a measurement on the present platform.

Table 8: Mean latency decomposition and analytical camera-input-to-LED budget (ms).

Detector setting	Capture	Detect	Track	Serial/LED	Estimated end-to-end
Main YOLO11n	6.66	239.00	1.50	2.39–5.30	250.77–253.68
Legacy YOLOv5n	6.60	146.32	1.66	2.39–5.30	158.42–161.33
Legacy YOLOv8	5.77	314.00	1.39	2.39–5.30	324.67–327.58
Legacy NMS ensemble	5.94	446.77	2.05	2.39–5.30	458.89–461.80
Legacy weighted ensemble	5.66	455.29	2.32	2.39–5.30	467.69–470.60

Table 9: Scenario-based deployment latency estimate (ms).

Deployment scenario	Camera	Detector	Track/ control	Serial+LED	Projected end-to-end
Current CPU with live-camera envelope	10–33.3 ^A	239.00 ^M	1.50 ^M	2.39–5.30 ^E	252.89–279.10
Accelerated-detector projection	10–33.3 ^A	15–25 ^A	1.50 ^M	2.39–5.30 ^E	28.89–65.10
Conservative 30-FPS budget	10.00 ^A	≤ 16.50	1.50 ^M	≤ 5.30	≤ 33.30

^MMeasured on the CPU benchmark; ^Aassumed deployment envelope; ^Eanalytical serial/LED estimate.

Under the current CPU detector, replacing offline decode with the live-camera envelope yields an estimated 252.89–279.10 ms camera-to-full-grid-update latency. If detector inference is reduced to 15–25 ms, the projected range is 28.89–65.10 ms. Meeting 33.3 ms under the conservative 10-ms camera and 5.30-ms serial/LED assumptions requires detector inference no greater than 16.50 ms. Queueing, operating-system scheduling, thermal throttling, and pipeline overlap are excluded; the table therefore defines a design target, not a production latency claim.

Relative to the legacy YOLOv5n single detector, NMS and weighted ensembling increase mean software latency by 192.6% and 198.2%, respectively, while reducing GSSR from 14.38% to 8.29% and 8.81%. Relative to the legacy YOLOv8 single detector, the corresponding latency penalties are 41.6% and 44.4%, again without a GSSR gain. Thus, neither ensemble lies on the observed GSSR–latency Pareto frontier. This result supports the final single-detector pipeline: in this protocol, running both networks adds substantial cost while failing to improve downstream actuation. The GSSR values in Table 7 are frame-weighted over all 953 frames, whereas Table 6 first computes each scenario result and then averages the five scenarios equally. The two tables answer different questions and their GSSR values should not be interchanged.

5.3 Main Control-Target Filtering Stress Evaluation (D0–D7)

The main primary-detector pipeline is evaluated through D0–D6 on four development scenarios; D7 replays frozen D6 predictions on the designated validation scenario at three resolutions. D7 is therefore a replay metric, not a direct runtime GSSR.

Table 10: Main primary-detector staged evaluation. Values are means over the four D0–D6 development scenarios or the designated D7 scenario and five fixed seeds; percentages are used unless otherwise stated.

Stage	Grid	GSSR / M-GSSR	SCR	FDR	Grid IoU	CmdChg
D0 runtime baseline	8×8	9.59	60.81	57.30	0.195	28.78
D4 filtered control interface	8×8	16.64	53.56	47.79	0.214	44.18
D5 filtered + depth-aware control	8×8	13.02	47.56	44.43	0.184	27.75
D6 filtered final runtime stage	8×8	13.02	47.98	44.56	0.187	11.29
D7 replay validation	8×8	44.49	53.22	57.14	0.429	–
D7 replay validation	16×16	41.37	48.02	65.51	0.345	–
D7 replay validation	32×32	33.89	38.46	72.05	0.280	–

Relative to D0, D4 combines adaptive noise, multi-cue association, and target filtering to raise runtime GSSR from 9.59% to 16.64% and reduce FDR from 57.30% to 47.79%. D6 retains the lower-FDR regime while reducing command changes to 11.29%. In replay, M-GSSR is highest at 8×8 (44.49%) and declines at finer grids. Table 11 isolates the effect of the control-target filter against the corresponding unfiltered pipeline.

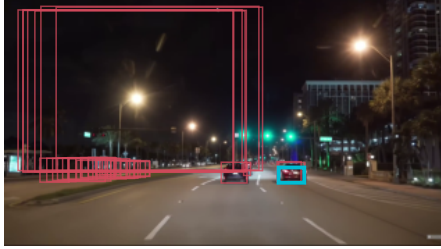
The largest per-scenario gain occurs in the street-lit night sequence: D4 GSSR rises from 2.94% to 27.35% while FDR falls from 82.57% to 56.02%; at D6, GSSR rises from 1.96% to 31.08% and FDR falls from 93.61% to 30.75%. Figure 5 shows a representative D6 frame from this sequence. The unfiltered command accepts 32 targets and suppresses 49 cells; the same tracker state after filtering passes two targets and exactly matches the two-cell ground-truth mask. This frame is illustrative rather than an aggregate result; the aggregate reductions are reported in Table 11.

Across the audit, high-FDR/low-IoU candidates fall from 4533 to 3038 for D4 and D6 combined (33.0%) and from 2296 to 1390 for D6 alone (39.5%). The filter remains conservative: D6 SCR falls from 63.53% to 47.98% and D5 SCR from 57.67% to 47.56%. It is therefore a lower-FDR baseline rather than an optimized policy; threshold sensitivity remains future work.

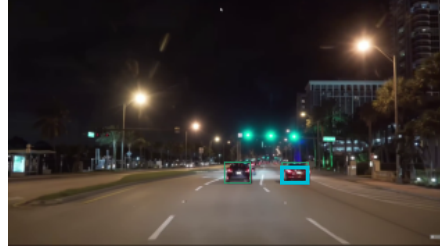
Table 11: Effect of control-target filtering in the staged M-GSSR evaluation. Values are percentages unless otherwise stated. The filtered column uses the control-target filter in Table 3.

Metric	Unfiltered	Filtered	Change
D4 GSSR	10.26	16.64	+6.38
D4 FDR	58.71	47.79	-10.92
D5 GSSR	5.06	13.02	+7.96
D5 FDR	64.63	44.43	-20.20
D6 main GSSR	5.85	13.02	+7.17
D6 main FDR	64.82	44.56	-20.25
D7 replay 8×8 M-GSSR	26.10	44.49	+18.39
D7 replay 8×8 FDR	73.07	57.14	-15.93

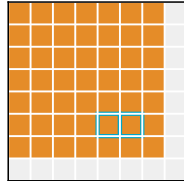
(a) Unfiltered control targets (32)



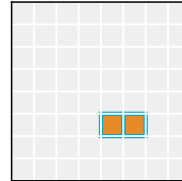
(b) Filtered control targets (2)



(c) Unfiltered MBOX: IoU 0.0408, FDR 0.9592



(d) Filtered MBOX: IoU 1.0000, FDR 0.0000



— Unfiltered target — Filtered target — Ground truth — Suppressed cell

Fig. 5: Representative control-target filtering result for D6 on the street-lit night sequence (seed 11, video frame 203). Red and green boxes are the unfiltered and filtered control targets, respectively; cyan denotes ground truth. The lower panels compare the resulting MBOX masks.

5.4 Comparison with SOTA Online Tracking Baselines

Table 12 compares the primary-detector pipeline with ByteTrack [16] and BoT-SORT [17] under the same five-scenario protocol. It is an actuation comparison, not a generic MOT benchmark, because annotations lack persistent IDs. The AuraBeam row uses the pre-filter KF3D output in this tracker-controlled comparison and should not be numerically interchanged with the D6 staged result.

Table 12: Comparison with SOTA online tracking baselines under the main detector protocol. Values are five-scenario means.

Tracker	GSSR	SCR	FDR	Grid IoU	Jitter (px)	FPS
Raw detector / no tracker	8.39	28.09	34.77	0.119	60.06	21.15
ByteTrack	9.09	23.90	29.60	0.122	44.59	21.43
BoT-SORT	8.36	21.85	22.93	0.104	37.80	13.09
AuraBeam KF3D (pre-filter)	7.31	53.71	60.84	0.138	55.79	14.22

ByteTrack has the highest mean GSSR and BoT-SORT the lowest FDR, but both have much lower SCR than AuraBeam (23.90% and 21.85% versus 53.71%). The pre-filter AuraBeam row also has the highest grid IoU, at the cost of conservative over-darkening; the main D0–D7 study targets this trade-off at the control interface.

5.5 Adaptive Observation: Controls and Targeted Degradation

Figure 6 summarizes the five-case roles and the B3/B4 actuation comparison.

Table 13: Negative controls: full-sequence actuation metrics (% , mean of five replays)

Scenario	Role	B3		B4		Shared MGR	B4 switches
		GSSR	FDR	GSSR	FDR		
Mountain curve at night	Negative control	84.21	15.79	84.21	15.79	0.00	2
Urban night with street lighting	Negative control	95.59	4.41	95.59	4.41	0.00	0

Five-scenario robustness evaluation protocol

Number inside marker = evaluated frames

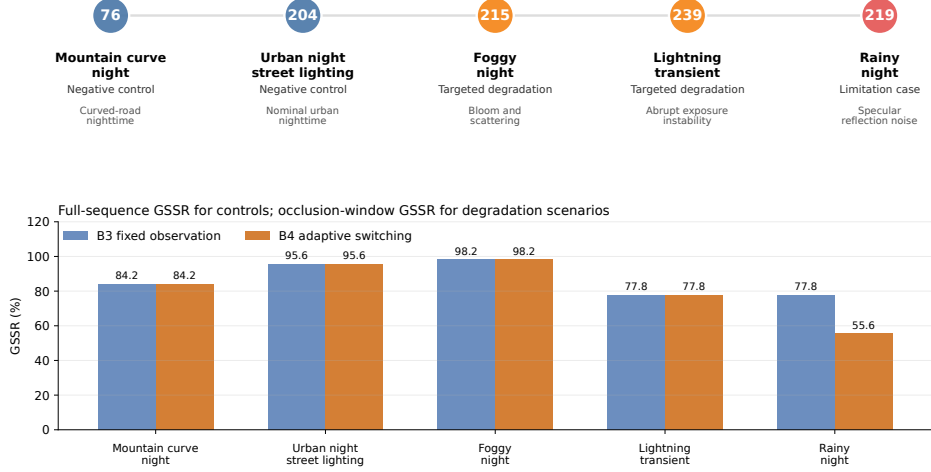


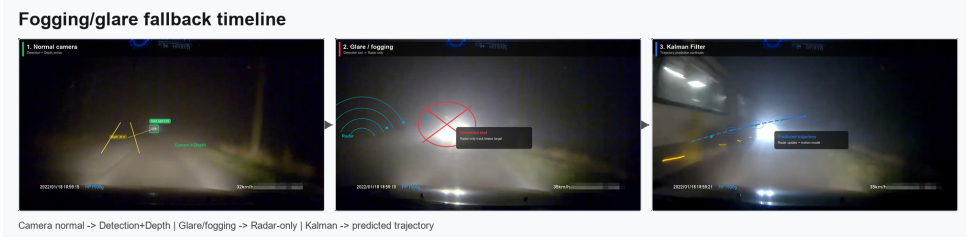
Fig. 6: Five-scenario robustness evaluation protocol. The upper panel summarizes the evaluated frame counts, scenario roles, and dominant visual challenges for the two negative controls, two targeted degradation scenarios, and the rainy-night limitation case. The lower panel compares B3 and B4 using full-sequence GSSR for the control scenarios and occlusion-window GSSR for the degradation and limitation scenarios.

The two controls show identical B3/B4 actuation metrics. B4 never switches in the urban sequence and switches only twice in the mountain sequence, so the controls support preservation but not superiority.

Table 14: Primary comparison on degradation intervals (% , mean of five replays)

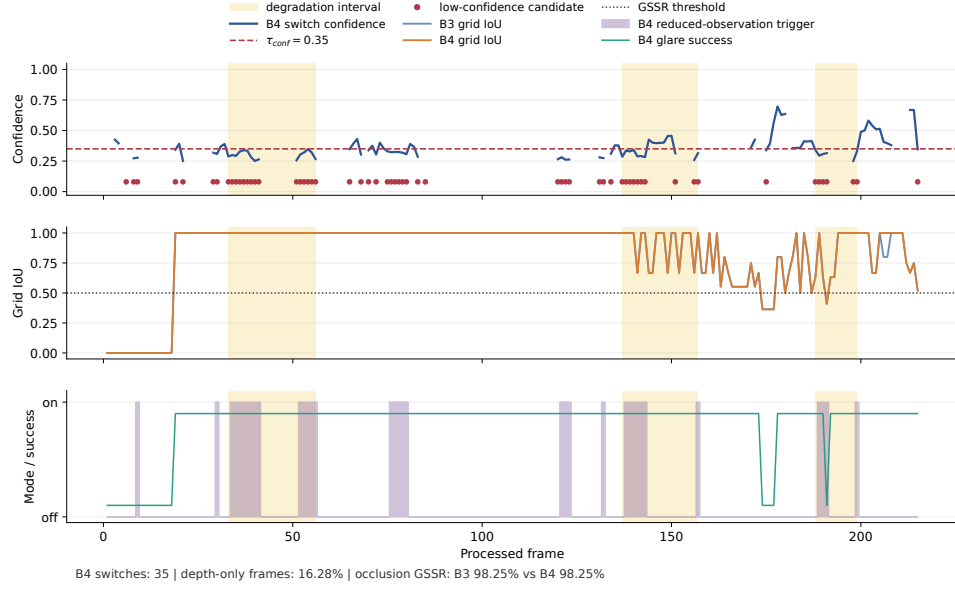
Scenario	Role	B3			B4			
		GSSR	MGR	FDR	GSSR	MGR	FDR	Switches
Foggy night	Targeted degradation	98.25	0.00	1.75	98.25	0.00	1.75	35
Lightning disturbance at night	Targeted degradation	77.78	0.00	22.22	77.78	0.00	22.22	1
Rainy night	Limitation case	77.78	0.00	22.22	55.56	0.00	44.44	9

Table 14 restricts metrics to annotated degradation intervals. B4 switches 35 times in fog and once in lightning, yet matches B3 in both cases. Thus the fallback can preserve selected degradation cases but does not establish a universal gain. Figure 7a illustrates the corresponding depth-only transition.



(a) Qualitative fallback sequence.

Foggy-night switching behavior



(b) Frame-level switching and actuation signals.

Fig. 7: Foggy-night adaptive fallback. The qualitative sequence shows the transition from full observation to depth-only propagation, while the timeline links confidence-triggered switching to LED-grid actuation quality.

Figure 7b reports the corresponding frame-level signals. The confidence trace shows when the switching guard becomes active, while the grid-IoU and glare-success traces verify that reduced-observation updates preserve the LED-grid outcome during the annotated windows.

Rainy-night limitation case

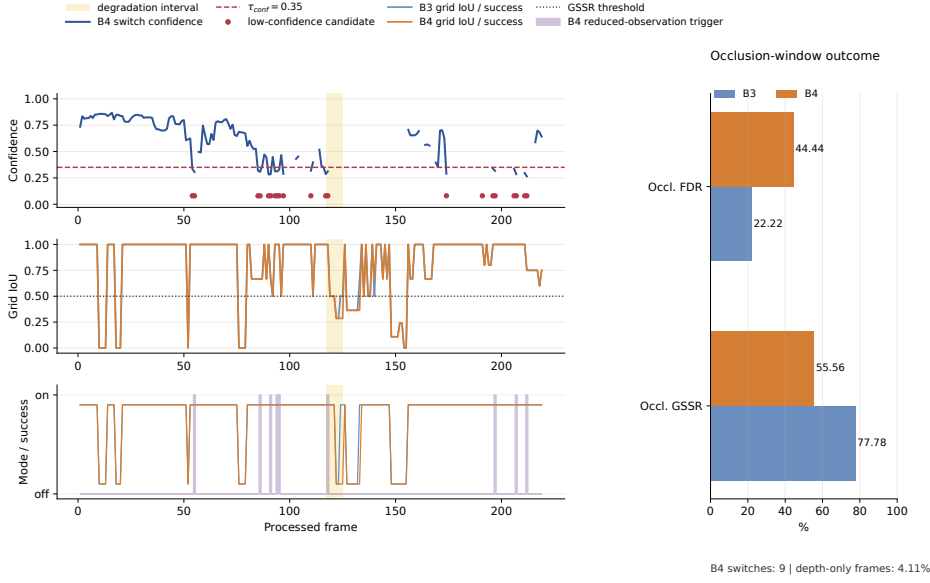


Fig. 8: Rainy-night limitation case. The time-series panels show B4 switch confidence, grid IoU, reduced-observation activations, and glare-suppression success during the rain sequence. The summary bars highlight the boundary of the adaptive policy: B4 reduces occlusion-window GSSR relative to B3 while increasing occlusion false darkening.

5.6 Adaptive Observation Limitation: Rain

Rain is the boundary case: B4 lowers occlusion GSSR from 77.78% to 55.56% and raises FDR from 22.22% to 44.44%, despite zero MGR for both methods. Nine switches per replay under specular noise suggest that confidence toggling disrupts command continuity. Figure 8 shows this failure. The archived evaluation-loop mean changes only modestly (277.65 to 282.12 ms), and RMSE differences are similarly small; these historical values include evaluator overhead and are not compared with the dedicated software control-path benchmark in Table 7. The actuation metrics provide the decisive evidence.

5.7 Discussions and Limitations

The results support robust tracking and command scheduling, but define a clear boundary for confidence-based switching. The primary checkpoint (7.31% GSSR, 60.84% FDR) outperforms the historical weighted ensemble (7.18%, 72.44%), yet per-scenario variation shows that detector quality alone is insufficient. Image-space gains can become false darkening after grid projection, motivating the complete perception-to-actuation evaluation.

The latency benchmark adds a computational boundary to this conclusion. Detector inference accounts for 93.8–97.8% of mean software-control latency, whereas tracking and mbox generation require only 1.39–2.32 ms. Both legacy ensembles are dominated by single-detector alternatives in the observed GSSR–latency plane, and all settings miss the 30-FPS deadline on the tested CPU. Serial transmission and a full LED refresh add an estimated 2.39–5.30 ms, so detector acceleration, reduced input resolution, or a GPU/edge accelerator is required before real-time claims can be made. Under the conservative deployment envelope in Table 9, detector inference must be reduced to at most 16.50 ms to satisfy the 33.3-ms deadline.

KF2D cuts centroid error by more than half, while KF3D adds a 20-point GSSR gain through depth-aware actuation. Hold-time reduces MGR from 1.20% to 0.37% with a small FDR increase (8.35% to 8.81%); its main benefit is continuity, not a large reduction in command changes.

Adaptive switching preserves the fog and lightning intervals but fails in rain: occlusion GSSR falls from 77.78% to 55.56% and FDR rises from 22.22% to 44.44%, while MGR remains zero. The failure is grid misalignment or over-darkening rather than complete target loss, confirming that image-space metrics alone would miss the relevant safety trade-off.

Control-target filtering reduces D6 FDR from 64.82% to 44.56% while raising GSSR from 5.85% to 13.02%; D7 8×8 M-GSSR rises from 26.10% to 44.49%. Because SCR also falls, the thresholds are a conservative correction rather than an optimum. Robustness is therefore determined by the full perception-to-actuation chain.

The study is limited to five curated scenarios and a low-cost prototype, so it does not cover the diversity of real traffic, weather, geometry, or multi-vehicle interaction. Pseudo-Radar is a physics-based prior initialized from a visual acquisition, not validation against physical radar or LiDAR; an immediately saturated target cannot initialize it. The 8×8 actuator and GSSR/M-GSSR measure geometric LED coverage rather than calibrated photometry or regulatory compliance. Finally, missing persistent IDs preclude IDF1. AuraBeam should therefore be read as an actuation-aware degradation study, not a production ADB system.

6 Conclusion

AuraBeam combines primary YOLO detection, a Pseudo-Radar depth prior, adaptive KF3D, and LED-grid actuation to study glare-induced observation loss in camera-centric ADB. Adaptive switching preserves the controls and selected fog/lightning intervals but degrades the rainy limitation case, so it is not a universal replacement for fixed observation. Control-target filtering reduces false darkening and raises D7 8×8 M-GSSR from 26.10% to 44.49%. The CPU benchmark further shows that ensembling adds substantial latency without improving GSSR and that the current prototype does not meet the 33.3-ms deadline; the projected deployment budget requires detector inference of at most 16.50 ms under the stated camera and actuator assumptions. Robust ADB behavior therefore depends on temporal estimation, target selection, and actuation-aware evaluation in addition to detector quality.

Appendix A Supporting Historical Ablations

The archived B/C experiments use the historical weighted detector and a legacy aggregation protocol. They are retained to document the design progression but are not numerically mixed with the main primary-detector D0–D7 study.

A.1 Tracking and Observation Configurations

Table A1: Historical tracking and observation ablations B1–B4. Values are means over five replays of the five-scenario set.

Tracking ID	configuration	RMSE	JR%	GSSR	FDR	MGR	Occl. GSSR	Occl. FDR
B1	Raw / direct	122.32	0.00	56.32%	6.69%	19.98%	22.61%	1.75%
B2	KF2D / direct	56.16	2.68	71.10%	5.80%	1.20%	48.15%	7.99%
B3	KF3D fixed / hold	56.16	2.68	91.10%	8.54%	0.37%	84.60%	15.40%
B4	KF3D adaptive / hold	56.49	3.03	90.82%	8.81%	0.37%	77.19%	22.81%

KF2D/direct reduces RMSE from 122.32 to 56.16 px and lowers MGR from 19.98% to 1.20%. KF3D/hold reaches 91.10% full-sequence GSSR, while the adaptive B4 variant preserves full-sequence behavior but lowers occlusion GSSR from 84.60% to 77.19%. Because B2 and B3 also differ in scheduling, their difference is not interpreted as a pure tracker ablation.

A.2 Command Scheduling and Switching

Table A2: Historical control and switching ablations C1–C4. Values are means over five replays of the five-scenario set.

Control ID	configuration	GSSR	FDR	MGR	Occl. GSSR	Occl. FDR	Occl. MGR	Switch count	CmdChg
C1	Direct / no hold	90.45%	8.35%	1.20%	77.19%	22.81%	0.00%	9.4	17.44%
C2	Hold-time only	90.82%	8.81%	0.37%	77.19%	22.81%	0.00%	9.4	17.26%
C3	KF3D fixed obs + hold	91.10%	8.54%	0.37%	84.60%	15.40%	0.00%	0	17.35%
C4	KF3D adaptive + hold	90.82%	8.81%	0.37%	77.19%	22.81%	0.00%	9.4	17.26%

Hold-time lowers MGR from 1.20% to 0.37% while slightly raising FDR from 8.35% to 8.81%. The command-change reduction is small, and fixed-observation C3 remains stronger on the occlusion-window aggregate.

A.3 Protocol-Level Direction Checks

Table A3: Paired direction checks for the supporting ablations. Values summarize deterministic scenario-run entries and do not represent independent statistical trials.

Comparison	Metric	Left mean	Right mean	Δ	Interpretation
Historical detector pair	GSSR $\uparrow$	43.51	56.32	+12.81	Legacy detector comparison.
Historical fusion pair	GSSR $\uparrow$	56.15	56.32	+0.17	Fusion effect is small.
B1 $\rightarrow$ B2	RMSE $\downarrow$	122.32	56.16	-66.16	KF2D reduces centroid error.
B2 $\rightarrow$ B3	GSSR $\uparrow$	71.10	91.10	+20.00	KF3D/hold gives the largest actuation gain.
C1 $\rightarrow$ C2	CmdChg $\downarrow$	17.44	17.26	-0.19	Hold-time slightly reduces changes.
B3 $\rightarrow$ B4	Occl. GSSR $\uparrow$	84.60	77.19	-7.41	Adaptive switching is worse in aggregate.
C3 $\rightarrow$ C4	CmdChg $\downarrow$	17.35	17.26	-0.09	Smoother commands do not improve GSSR.

These checks summarize effect direction only. The repeated replays are fixed offline executions of the same scenarios, so inferential significance testing is not used.

Declarations

The authors declare no conflict of interest, particularly no financial and personal relationships with other people or organizations that could inappropriately influence (bias) this work.

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